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TrustEstate

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1. Executive Summary

1.1 Project Overview

TrustEstate is a comprehensive web-based real estate management platform designed to modernize and streamline the lifecycle of property transactions. Traditional real estate processes often rely on fragmented communication, manual documentation, and limited transparency between stakeholders such as property owners, buyers, agents, and inspectors. These inefficiencies can lead to delays, miscommunication, and reduced trust in the transaction process.

TrustEstate addresses these challenges by providing a centralized, role-based digital platform that integrates property listing management, offer negotiation, agent mediation, and structured inspection workflows into a single system. The platform ensures that each participant in the real estate process operates within clearly defined responsibilities, improving accountability and reducing operational friction.

The system introduces a multi-role architecture consisting of Buyers/Tenants, Property Owners, Agents, Property Inspectors, and Administrators. Each role interacts with the system through a tailored interface, enabling efficient collaboration while maintaining strict access control and data security.

Key features of TrustEstate include advanced property search and filtering, structured offer negotiation with multiple rounds, agent-managed listing approval, inspection scheduling and reporting, and real-time notifications across all stages of the transaction. By digitizing and automating these workflows, the platform enhances transparency, improves communication, and builds trust among all involved parties.

Accessible from any device with internet connectivity, TrustEstate offers a user-friendly and responsive interface that supports seamless interaction between users. The system ultimately aims to increase efficiency, reduce manual overhead, and provide a reliable and transparent real estate experience.

1.2 Purpose and Scope of this Specification

Purpose

This specification document defines the functional and non-functional requirements of the TrustEstate platform. It serves as a comprehensive guide for developers, stakeholders, and evaluators to understand the system’s architecture, features, and operational behavior.
The purpose of this project is to design and develop a robust real estate management system that digitalizes and automates property transactions, from listing creation to final deal closure. The platform facilitates structured collaboration between multiple user roles while ensuring data integrity, security, and transparency throughout the process.
Additionally, this document aims to establish a clear understanding of system functionalities, including user role permissions, workflow processes, and interaction mechanisms. It also supports effective communication within the development team, ensuring alignment with project goals and requirements.

Scope:

In Scope

The TrustEstate platform includes the following core functionalities:

- Property Listing Management: Enables Property Owners to create and manage listings, while Agents review, approve, and publish them.
- Role-Based Access Control: Implements distinct permissions and workflows for Buyers/Tenants, Property Owners, Agents, Inspectors, and Admins.
- Advanced Search and Filtering: Allows users to browse and filter listings based on location, price, property type, and listing type.
- Offer and Negotiation System: Supports structured offer submission, counter-offers, and up to three negotiation rounds managed by Agents.
- Inspection Management: Facilitates scheduling, execution, and reporting of property inspections with standardized evaluation criteria.
- Real-Time Notifications and Messaging: Keeps users informed about status changes, offers, inspections, and system events while enabling controlled communication between parties.
- Administrative Control and Monitoring: Provides Admins with tools to verify users, manage disputes, monitor listings, and oversee system activity.

Out of Scope

The following aspects are not included in the current scope of the TrustEstate system:

- Integration with external payment gateways or financial transaction processing systems.
- Legal contract generation or digital signing of property agreements.
- Mobile-native applications (the system is web-based, though responsive).
- Hardware-related features such as IoT-based property monitoring systems.
- Post-deployment maintenance and support services, which will be defined separately.

2. Product / Service Description

2.1 Product Context

TrustEstate is a standalone, self-contained web-based platform that does not depend on or integrate with any external real estate systems or third-party listing services. All core workflows - listing management, offer negotiation, inspection scheduling, and administrative oversight - are handled entirely within the platform. The system interfaces internally between five distinct user roles: Buyers/Tenants, Property Owners, Agents, Property Inspectors, and Administrators, each interacting through a role-specific interface within the same environment. While the platform operates independently, it relies on a hosting infrastructure for deployment and availability, and on an email/notification service to deliver real-time alerts to users across all transaction stages.

2.2 User Characteristics

Buyer / Tenant (Customer)

- Represents the end consumer of property listings, seeking to purchase or rent a property through the platform.
- Self-registers freely through the public registration page with no professional type selection; account is activated immediately upon registration with no Admin approval required.
- Browses and filters listings by location, price range, property type, and listing type (sale/rent) without requiring a login.
- Registers and logs in to access full platform features, including saving favorites, submitting offers, and messaging Agents.
- Submits offers on active listings with a proposed price and optional message, and participates in up to 3 rounds of negotiation per offer.
- Receives real-time notifications at every key status change relevant to their offer or transaction, including offer responses and inspection outcomes.
- Views completed inspection reports for listings on which they have an accepted offer, and may withdraw or revise their offer following inspection verdicts.
- Requires basic familiarity with web-based platforms and online transaction processes.

Property Owner

- Represents individuals or entities seeking to list their property for sale or rent through the platform.
- Self-registers through the public registration page by selecting Property Owner as their professional type; account is activated immediately with no Admin approval required.
- Creates and submits property listings - including title, description, location, asking price, listing type, and photos - and assigns an available verified Agent to manage the listing.
- Edits listing details while in Pending Agent Review status or after an Agent has requested corrections, but cannot publish listings directly.
- Monitors all their listings and current statuses from their owner dashboard, and receives notifications at every relevant stage of the transaction.
- Views completed inspection reports for their properties and communicates with the assigned Agent regarding listing or transaction matters.
- Cannot accept, decline, or counter offers, as these actions are handled exclusively by the assigned Agent.
- Requires moderate familiarity with property listing processes and digital platforms.

Agent

- Acts as the primary intermediary between Property Owners and Buyers/Tenants, managing listings, negotiations, and inspection coordination.
- Self-registers through the public registration page by selecting Agent as their professional type and providing agency affiliation details; account is created in Pending Verification status and requires Admin approval before activation.
- Upon approval, receives a Verified badge on their profile, which is visible to Property Owners when selecting an Agent for a listing assignment.
- Reviews submitted listings, requests corrections if needed, and approves listings to set them live on the platform.
- Manages the full offer lifecycle on behalf of the Property Owner - accepting, declining, or countering offers, and handling up to 3 rounds of negotiation per offer.
- Assigns verified Property Inspectors to listings with accepted offers, reassigns Inspectors if needed, and updates listing status throughout the transaction lifecycle.
- Marks transactions as closed once all conditions are met, archiving the listing and notifying all relevant parties.
- Requires professional knowledge of real estate transactions, strong communication skills, and proficiency with digital management tools.

Property Inspector

- Responsible for conducting structured physical inspections of properties with accepted offers and submitting formal inspection reports through the platform.
- Self-registers through the public registration page by selecting Property Inspector as their professional type; account is created in Pending Verification status and requires Admin approval before activation.
- Upon approval, receives a Verified badge on their profile, which is required for assignment eligibility by Agents.
- Views all inspections assigned to them, including property details, scheduled inspection date, and the assigned Agent's contact information.
- Submits structured inspection reports organized into four fixed categories - Structural Integrity, Plumbing, Electrical, and Safety - with findings, pass/fail ratings, severity classifications (Minor, Moderate, Critical), and supporting photos per category.
- Assigns a final verdict of Passed, Passed with Conditions, or Failed, triggering automatic notifications to the Buyer/Tenant, Agent, and Property Owner upon submission.
- Cannot modify or resubmit a report once formally submitted, and does not communicate directly with Buyers/Tenants or Property Owners outside of the report.
- Requires professional inspection qualifications, attention to detail, and familiarity with structured digital reporting tools.

Admin

- Oversees platform integrity, user verification, content compliance, and dispute resolution across the entire system.
- Pre-seeded into the database at deployment time; no registration or signup flow exists for this role.
- Reviews and approves or rejects pending Agent and Property Inspector accounts within 72 hours of submission, notifying users of the outcome.
- Monitors all active listings on the platform, and may flag, suspend, or remove listings that violate platform policies with a documented reason.
- Deactivates, suspends, or reactivates user accounts as necessary, notifying affected users with documented reasons.
- Manages disputes submitted by any user by reviewing listing details, offer history, inspection reports, and communications, and may escalate disputes by temporarily suspending a transaction if one party is unresponsive.
- Accesses a real-time analytics dashboard displaying active listings, flagged listings, pending verifications, ongoing transactions, registered users per role, and open disputes.
- Cannot create accounts for any user, approve or reject individual listings, or act on behalf of any other role within their specific workflows.
- Requires strong administrative judgment, familiarity with platform governance processes, and proficiency with dashboard-based monitoring tools.

2.3 Assumptions

- All users are assumed to have basic familiarity with web-based platforms and standard online interaction patterns.
- The system operates under the assumption of a stable and reliable internet connection for all parties involved.
- Property Owners are responsible for providing accurate and complete listing information at the time of submission.
- Agents are assumed to operate in good faith on behalf of Property Owners within the boundaries defined by their role.
- Property Inspectors are assumed to hold the professional qualifications necessary to conduct property evaluations in compliance with relevant standards.
- The Admin will complete all pending verification reviews within the defined 72-hour window.
- Listing photos and supporting documents uploaded by Property Owners and Inspectors are assumed to be authentic and accurate.
- All users are responsible for maintaining the confidentiality of their login credentials.
- Notifications sent by the system are assumed to reach users in a timely manner through the platform's notification mechanism.
- The platform assumes that no more than one verified Agent is assigned to manage any single listing at a given time.
- It is assumed that the system will perform regular data backups to prevent data loss in case of system failure or unexpected issues.

2.4 Constraints

- The platform is web-based and requires a stable internet connection for all operations; offline functionality is not supported.
- The system must comply with applicable data protection regulations, including GDPR, to ensure user data privacy and security.
- Role-based access control is strictly enforced; users cannot perform actions outside the permissions defined for their role.
- Agent and Property Inspector accounts cannot be activated without Admin approval, introducing a mandatory verification delay of up to 72 hours.
- All listings must pass Agent review before going live; Property Owners have no ability to publish listings directly.
- Offer negotiation is limited to a maximum of 3 rounds per offer; offers not resolved within this limit are expired and the listing returns to Active status.
- Inspection reports are immutable once submitted; Inspectors cannot modify or resubmit a completed report.
- The platform does not support integration with external payment gateways or legal document generation in its current scope.
- The user interface and functional behavior may be constrained by the limitations of the web technologies and frameworks used during development.

- Budget and timeline constraints during development may limit the implementation of certain advanced features or optimizations.
- The system is not a mobile-native application; while the interface is responsive, it is designed and delivered as a web-based platform.

2.5 Dependencies

- The activation of Agent and Property Inspector accounts is dependent on the Admin completing verification within the defined timeframe.
- The publication of a listing is dependent on the assigned Agent reviewing and approving it; no listing can go live without Agent action.
- The submission of an offer is dependent on at least one active listing being available on the platform.
- The assignment of a Property Inspector to an inspection is dependent on the Agent taking action after an offer has been accepted.
- The generation of an inspection report is dependent on the Inspector completing the physical inspection and submitting findings through the platform.
- Notifications to all relevant parties at each transaction stage are dependent on the platform's notification service operating correctly.
- The availability of listing data for browsing and filtering is dependent on the accuracy and completeness of the information submitted by Property Owners.
- The dispute resolution process is dependent on the Admin having access to all relevant listing details, offer history, inspection reports, and communications between parties.
- The platform's overall performance and availability are dependent on the reliability of the chosen hosting infrastructure.
- The development of the project is reliant on specific technologies for backend, frontend, and database management, as defined in the technical architecture.